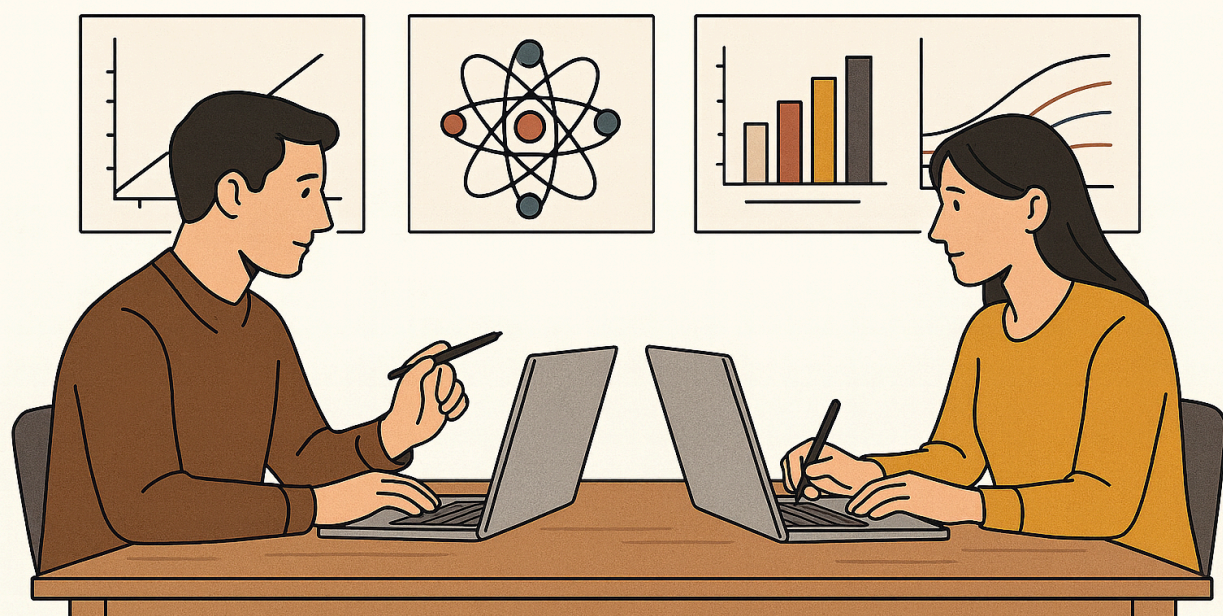


Welcome to the ALD/E-ImageMiner Annotation Task



ALD/E-ImageMiner Annotation Task – Guidelines

Welcome to the **ALD/E-ImageMiner Annotation Task**. We are glad you signed up and hope you find this experience enriching, fun, and stimulating to your knowledge as well.

Overview

From a total of 205 articles, each annotator will be allocated 25 scientific articles on the research topics of atomic layer deposition (ALD) and atomic layer etching (ALE). These scientific articles are from the broader field of materials science and engineering. The annotation task is defined based on the **figures** extracted from each of these articles. You are welcome to browse the complete collection of papers in our GitHub repository here:

👉 [Annotation Task Papers](#)

The work will be carried out using a simple and easy-to-use UI annotation tool that we have developed in-house and is also hosted on our servers. Each user will be assigned **25 papers**, with each paper consisting of one or more figures to be annotated. The detailed workflow and task-specific steps are presented in the following sections.

The overall goal of this pilot is to **curate and correct figure annotations** that were originally generated by the Qwen-2.5-VL-7B model. In some cases, the model annotations may already be correct, requiring no further action from you. In other cases, the model may be partially or completely wrong, and you will need to correct or recreate the annotation. For simple cases, we expect that the auto-generated annotations will already be accurate, saving you time. The real value of this project lies in the **human-corrected annotations** where the model falls short.

Using the Annotation Interface

The ALD-E Image Miner annotation tool consists of separate views to facilitate the user in a simple and easy-to-follow workflow as discussed below:

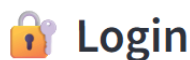
1. Login View



Deploy



ALD-E ImageMiner Annotation Tool



Login

Email

Login

As a first step, the annotator needs to enter their email which you use to communicate with us. We activate your account with just your email address.

2. Main View



ALD-E ImageMiner Annotation Tool

All PDFs

Paper 1

Paper 4

Paper 7

Paper 10

Paper 13

Paper 16

Paper 19

Paper 22

Paper 25

Logout

Paper 2

Paper 5

Paper 8

Paper 11

Paper 14

Paper 17

Paper 20

Paper 23

Paper 3

Paper 6

Paper 9

Paper 12

Paper 15

Paper 18

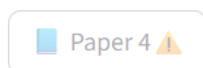
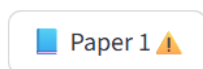
Paper 21

Paper 24

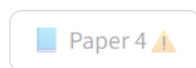
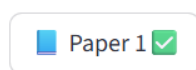
Next you will see the “main view” where you will see all your assigned papers. The Page shows a grid of buttons by clicking on which will navigate you to the “pdf view”.

- An "exclamation mark" on the button shows that the paper has not been completed yet. Furthermore, a disabled button means that the paper is not available for annotation to you yet. This is a feature which will be enabled once you have gone through the initial phase of annotating one paper and would like

to continue with the rest of the annotation task.



- Once a paper has been marked as completed, the icon on the associated button will change as shown below:



- A user can log out of the session by clicking on the “Logout” button at the bottom of the “main” view as shown below.

⚠ Attention: Please make sure that any changes you have made have been saved using the “Save” button. Otherwise, the changes will be lost.

3. Paper View

PDF View

Paper 1

Main Category

atomic-layer-etching

Sub Category

experimental-usecase

[View PDF \(Github\)](#)

Figures (7)

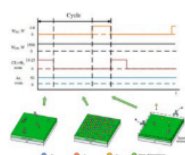


Figure 1 ⚠

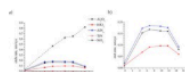


Figure 4 ✓

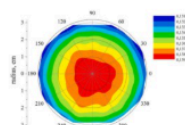


Figure 7 ⚠

☐ Completed

[Save](#)

[Back to All PDFs](#)

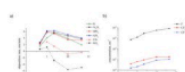


Figure 2 ⚠

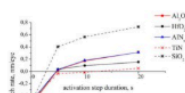


Figure 5 ⚠

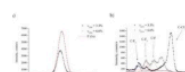


Figure 3 ⚠

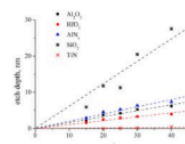


Figure 6 ⚠

The “Paper view” is the place where you have access to all the data for each research paper you have been assigned.

- At the top of the page are displayed the main research category and sub category of the paper.

Paper 1

Main Category

atomic-layer-etching

Sub Category

experimental-usecase

 View PDF (Github)

- To view the actual research article, you have to click on the “View PDF (Github)” button. This opens a new browser tab so your main view will not be affected.
- The section “Figures” shows all the figures in a grid, which are in this research article. As shown below, you will see a thumbnail preview of the figure below which is a button. Click the button to open the “annotation view” for the figure. You will be able to perform the annotation task for that figure on the “annotation view page”.

Figures (7) ⇄

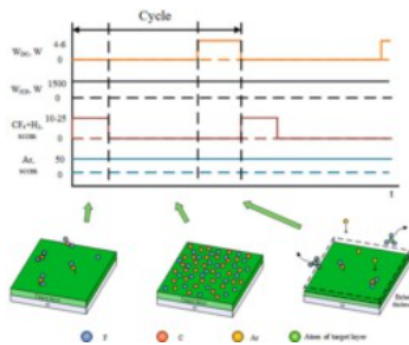


Figure 1 

- Visit the main paper page to visit each figure. After you have completed the annotation for a figure, on the main paper page, the button under the figure will reflect based on a changed icon whether it has been marked as “completed”, as shown below.

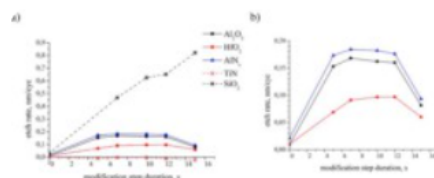


Figure 4 

- A paper can be marked as “completed” once a user has reviewed the changes, to track the progress through the annotation task as shown below.

☒ Completed

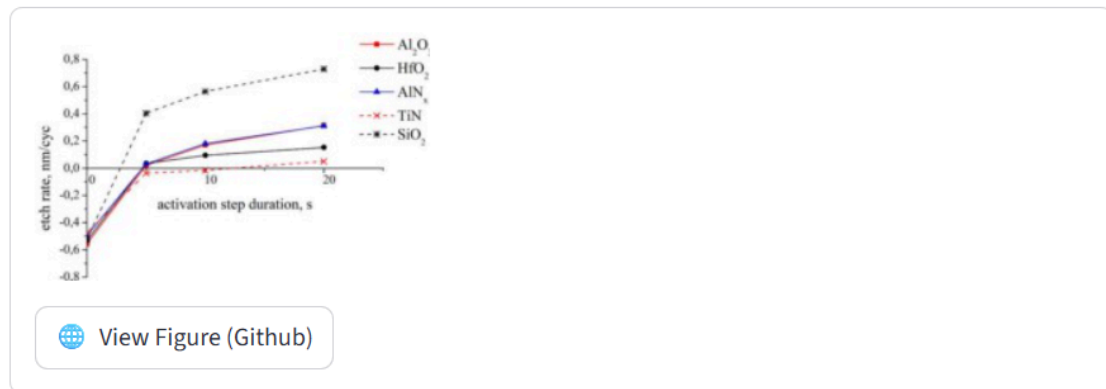
 Save

A user can still revisit figures marked as completed, to make edits or revise their previous annotations any number of times.

4. Figure Annotation View

Annotation View

Figure 5



The figure “annotation view” is the screen where a user annotates a figure for the following four tasks: 1) classification, 2) data table curation, 3) figure content summarization and 4) Q&A over the figure. Each of these tasks are defined later in these guidelines.

- The user can view the original extracted figure from the article if they click on the “View Figure (Github)” button. The extracted figure opens in a window external to the current screen.
- Figures usually have a “caption” associated with them in a research article. This caption is displayed in the “Figure caption” field in non-editable mode. All content that is non-editable will be displayed in a grayed box

Important pointers for annotators:

- a. Users should save their work from time to time to make sure they are not logged out. This is one of the technical limitations of the development technology we’ve used. According to the official documentation, a user session exists as long as the browser tab is open and connected to the service. We are looking for ways to overcome this to auto-save but the feature is not implemented yet.
- b. Users should also review their earlier “completed” annotations, for example, after finishing each PDF. As you progress and gain experience, your annotation decisions will become more consistent. Therefore, if you notice inconsistencies in your previous

annotations based on your improved understanding, you are expected to revise them accordingly.

The annotation tool is simple and easy-to-use, while still giving you easy access to the papers and figures you will be working on.

Next we describe the four annotation tasks that must be performed as part of this annotation endeavor.

The Four Annotation Tasks

You may not need to do all four tasks for every figure — the rules below explain when to stop.

1. Figure Classification

- This task relies on a **taxonomy of 40 figure types**, including an “*unknown*” category.
 - See the [full list here](#), which includes brief descriptions, parent taxonomy, and alternative names.
- A figure may be composed of **multiple subfigures**.
 - Read left to right, top to bottom, and follow the *a, b, c* ... labeling scheme if subfigures are not already labeled.
- The following pane on the annotation screen depicts the Figure classification task.

Classification

Auto Class

a line chart

Figure Classes

area chart

+ Append

Manual Class

Comment on Auto Class

- The “Auto Class” field is prepopulated with the classification performed automatically using a vision LLM. The small button on the bottom left can be used to copy the data in

the field to the “clipboard” which might be helpful when editing the “Manual Class” field.

- After verification of the figure, a user can update the “Manual Class” field by first selecting a “Figure Class” from the dropdown list and then clicking the “+ Append” button as shown below. This is due to the fact that sometimes figures contain sub-figures and the automated class might not have identified them. So a user needs to add them one by one here by. This step has to be repeated for all the sub-figures.

The screenshot shows a web interface with two main sections. The top section, titled 'Figure Classes', contains a dropdown menu with 'area chart' selected and a '+ Append' button. The bottom section, titled 'Manual Class', contains a large, empty text input field with a small edit icon in the bottom right corner.

- Users will have to add as many sub-figures as there needs to be. The format should be as “label TAB class”, separated by new lines. The user should update "label" should be alphabetical and in ascending order, i.e. a, b, c ..., as shown below:

This screenshot shows the same interface as the previous one, but the 'Manual Class' text field now contains two lines of text: 'a line chart' and 'label area chart', demonstrating the 'label TAB class' format.

⚠ Reminder: a label and a class should always be tab-separated.

- So there are two ways to do edit manual class field:
 - Copying the data from “Auto Class: field and fixing it based on the dropdown list.
 - Selecting via the dropdown list, hit append, and fix the label.
- The user should enter brief comments in the “Comment on Auto Class” field. This helps indicate whether the automatic classification was correct or if the user had to adjust it in the “Manual Class” field.

- **Next step rule:**

- If a figure contains mixed subfigures (some quantitative, some not):
 - Enter “a unknown” in the “Manual Class” field
 - Fill the “Comment on Auto Class” field describing the reason.
 - Mark the figure annotation as “completed”.
 - Save the changes
 - Click the “Back to PDF view” button

- 💡 *Tip:* You may use ChatGPT to assist with classification or justification (e.g., upload the figure and taxonomy and ask for candidate labels).
 - However, the **final decision is yours**. AI tools can only help check your reasoning.
 - Using AI to *perform* the task defeats the purpose of this project.

2. Data Table Curation

- This annotation project includes only **quantitative plots**, and every such plot is based on underlying numeric data.
- The goal of this task is to curate that numeric data in a usable form.
- The following fields belongs to the Data Table curation task:

Data

Auto Data

```

a
```markdown
| activation step duration, s | etch rate, nm/cyc |
|-----|-----|

```

📄

Manual Data

🌐 Online Markdown Editor

📄

- The “Auto Data” field represents data extracted by our automated process. The small button on the bottom left can be used to copy the data in the field to the “clipboard” which might be helpful when editing the “Manual Data” field.

Auto Data

```

a
```markdown
| activation step duration, s | etch rate, nm/cyc |
|-----|-----|

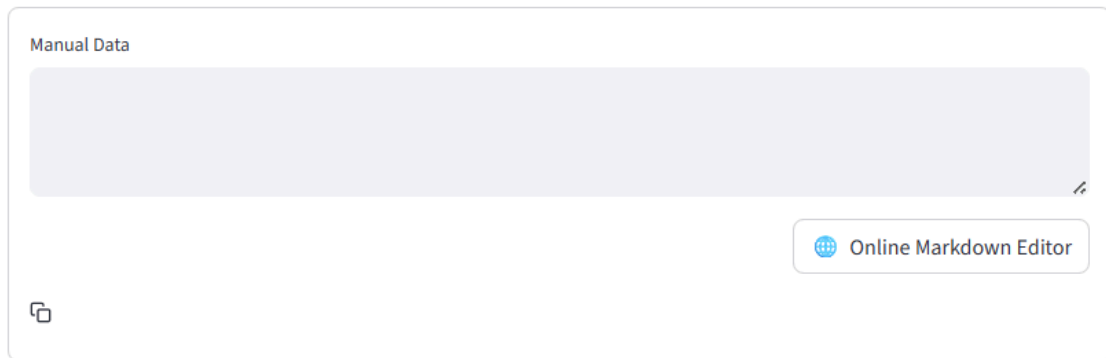
```

📄

In many cases, this auto data will be a good starting point. The data can be copied as usual using the “copy to clipboard” button below the field as shown above.

- Users will have to add as many data tables as there are sub-figures. Data tables are in standard [markdown table format](#). However, the label format itself should be as “label TAB table”, separated by new lines. Labels should be alphabetical and in ascending

order, i.e. a, b, c ..., as shown below:



Manual Data

Online Markdown Editor

- An external online markdown editor can be opened to help with correct markdown formatting by clicking on the button as shown below:



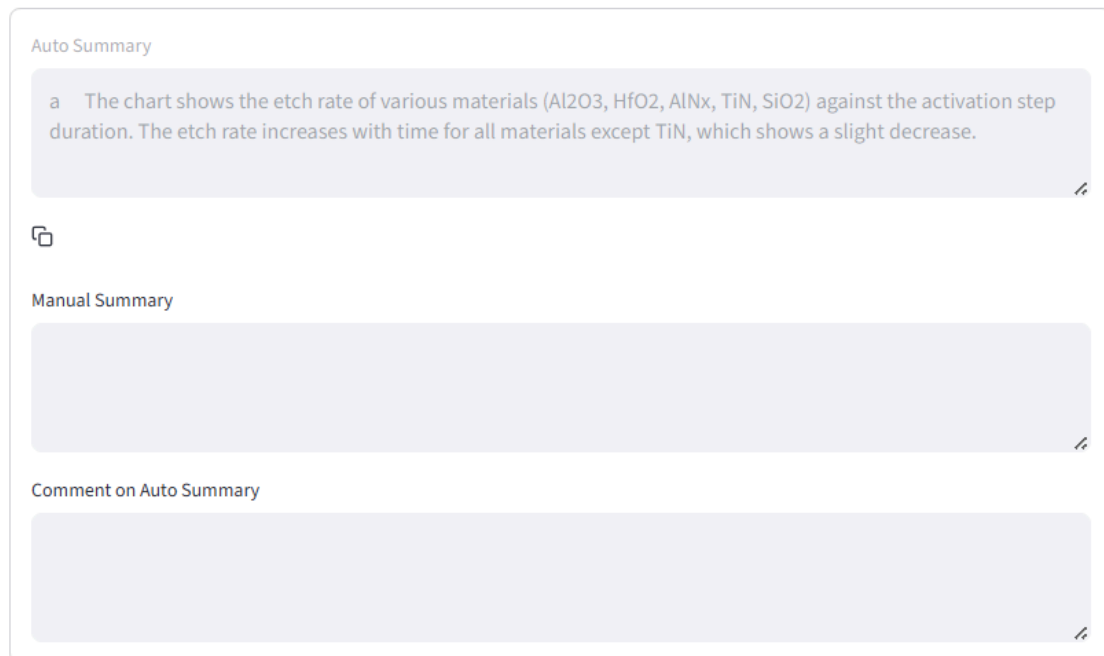
Online Markdown Editor

- Guidelines:
 - Do not remove newlines → each data point should remain on a separate line.
 - You do not need to annotate every single axis point.
 - Instead, capture only the salient points that help minimally reconstruct the plot.
- Example: Sub-figure data annotations should look like this: [figure_9.data.txt](#).

3. Figure Content Summary Curation

- The following fields belongs to the Figure Content Summary curation task:

Summaries



Auto Summary

a The chart shows the etch rate of various materials (Al₂O₃, HfO₂, AlN_x, TiN, SiO₂) against the activation step duration. The etch rate increases with time for all materials except TiN, which shows a slight decrease.

Manual Summary

Comment on Auto Summary

- The “Auto Summary” field contains a summary extracted by our automated process, based on an AI model Qwen, about the content of the figure or its subfigures.
- Your task is to validate this summary yourself.

- In many cases, Qwen’s summaries are too vague, generic, or repetitive and do not fully capture the figure content.
- If the auto summary is acceptable → leave it as is.
- If it is not acceptable →
 - Write the corrected version in “**Manual summary**” field.
 - Provide a brief justification in “**Comment on auto summary**” field explaining why the auto version was wrong.

4. Figure Q&A Annotation

- The following pane belong to the Figure Q&A annotation task:

Questions / Answers ↔

The screenshot shows a web-based form for Figure Q&A annotation. At the top, it says 'Q1/A1'. Below this, there is a 'Q1 Type' label followed by a dropdown menu currently set to 'Process-Oriented'. Underneath is a large, empty text box for 'Q1'. Below the 'Q1' box is an 'A1 Type' label followed by a dropdown menu currently set to 'Yes/No'. At the bottom is another large, empty text box for 'A1'.

- This task requires you to create **4 question–answer (Q&A) pairs per figure**.
Question types (choose one per question):
 1. Process-Oriented
 2. Comparative/Trend
 3. Structure–Property
 4. Application/Performance**Answer types (choose one per answer):**
 1. Yes/No
 2. Factoid
 3. List
 4. Paragraph (≥3 sentences)
- For detailed definitions and examples of these types, we ask you to read the following documentation: [Q&A README](#)
- ⚠ Please read the [Q&A README](#) file carefully before starting, and keep it at hand while annotating.
- The following is an example for the 1st Q&A:

- **Q1 type** → select the question type.

Q1 Type

Process-Oriented

- **Q1** → write your question.

Q1

- **A1 type** → select the answer type.

A1 Type

Yes/No

- **A1** → write your answer.

A1

- Repeat for Q2, Q3, and Q4.
- **Guidelines:**
 1. Each figure must have **4 Q&As**.
 2. At least **two-thirds of figures in a paper** must include at least one **paragraph-style answer** (≥3 sentences).
 - Example: For 6 figures, at least 4 must include paragraph answers.
 3. No more than **one-third of answers** in a paper may be **yes/no style**.
 - Example: For 6 figures, at most 2 may be yes/no.
 4. Avoid skewing: use a **balanced mix** of question and answer types.
 5. You are free to choose combinations of question and answer types, as long as these balance rules are met.

AI use:

- **✗ Do not use AI to generate questions or answers.** Both must be your own original work.
 - Part of the annotator compensation is specifically for the time and effort required to write original questions and answers based on your own ideas, creativity, expertise, and impressions from the images.
 - Our team is already well equipped to run vision–language models (VLMs) to generate Q&As, so this task has a *different purpose*: to capture authentic **human annotations** that cannot simply be reproduced by machines.
- **✓ You may use AI in a minimally non-invasive way only.**
 - Acceptable uses include double-checking linguistic aspects such as grammar, spelling, verbosity, or style.
 - AI should act only as a language helper, never as the source of scientific content or reasoning.

👉 The value of this project lies in your **human judgment, creativity, and unique perspective** when interpreting the images — not in outputs that machines could already generate.

Completion, Next Steps, and Contact

Annotation **quality** is the main criterion for selecting annotators after the pilot phase. We will evaluate your work based on careful attention to the instructions, correctness of classifications, curated data, summaries, and Q&As, as well as adherence to the balance rules (e.g., paragraph requirements and limits on yes/no answers). In short, we are looking for thoughtful, consistent, and reliable annotations that show care and understanding of the figures.

The long-term goal is to build a dataset of **205 fully annotated papers**, creating a valuable research resource for the community. By participating, you are directly contributing to advancing machine understanding of scientific images, with an impact on future AI and materials science research.

Once you have read these instructions carefully, please email us directly to request your annotation credentials:

- **fahad.ahmed [at] tib.eu**
- **jennifer.dsouza [at] tib.eu**

Before beginning with the annotation task, we will hold one-to-one onboarding sessions to address any questions about the annotation task or the annotation interface. We will respond with appointment times for the onboarding. We look forward to working with you and deeply value your contribution to this exciting project.

THANK YOU!

